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MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 51 MOD 1

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 51 MOD 1 impulse cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-D-21625 | Design and Evaluation of Cartridges for Cartridge Actuated Devices |
| MIL-P-116 | Preservation, Methods of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |

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| | |
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| MIL-STD-414 | Sampling Procedures and Tables for Inspection by Variables for Percent Defective |
| MIL-STD-129 | Marking for Shipment and Storage |

DRAWINGS

Naval Air Systems Command (Code Ident No. 30003)

| | |
|---|---|
| DL 2838297 | Cartridge, Impulse, MARK 51 MOD 1 |
| LD 534542 | Container, Inner, for Cartridge, Impulse, MARK 51 Type (Formerly Code Ident No. 10001) |
| LD 537905 | Container, Outer, for Cartridges, Power Source (Formerly Code Ident No. 10001) |
| 1558439 | Ignition Charge (Formerly Code Ident No. 10001) |
| 1825934 | Igniter Assembly (Formerly Code Ident No. 10001) |
| 1825935 | Bridgewire Assembly (Formerly Code Ident No. 10001) |
| 1825936 | Inner Cup and Electrode Assembly (Formerly Code Ident No. 10001) |
| 2838297 | Cartridge, Impulse, MARK 51 MOD 1, Assembly |
| 2171790 | Filler Plug (Formerly Code Ident No. 10001) |
| 2838286 | Loaded Spacer |
| 2838287 | Ignition Composition |
| 2838296 | Ignition Element Assembly |



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PUBLICATIONS

| | |
|---|---|
| BUSANDA Instruction 4030.17 | Preservation, Packaging, Packing and Marking; Navy Policy for |
| Bureau of Naval Weapons | |
| OP 5 | Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.) |
| WR-43 | Preparation of Quality Assurance Provisions |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

| | |
|---|---|
| 49 CFR 71-78 | Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

## 3. REQUIREMENTS

### 3.1 General.

3.1.1 The MARK 51 MOD 1 impulse cartridge covered by this specification shall be manufactured in accordance with DL 2838297 and all documents listed thereon.



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3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.3 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 76 MARK 51 MOD 1 impulse cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.4 Primary components. For the purpose of this specification, the Ignition Element, Drawing 2838296, is defined as a primary component. See 4.3.3.

3.5 Cleanliness and condition of parts. All parts shall be free of chips, dirt, grease, rust and all other foreign materials. There shall be no burrs nor other imperfections of manufacture which would harmfully affect either the assembly or the functioning of the product. The cleaning methods which are used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents which are used.

3.6 Ignition element. The ignition element used in the MARK 51 MOD 1 impulse cartridge shall be in accordance with Drawing 2838296 and shall have been manufactured within the preceding 12 months.

3.6.1 Explosive components, volatiles. The Ignition Composition, Drawing 2838287, shall have a maximum moisture content of 0.5% at time of loading as determined in 4.7.1. After consolidation of the ignition composition in the Loaded Spacer, Drawing 2838286, the loaded spacer shall be dried at +100°F to +120°F for a minimum of 16 hours. After removal of the units from the drying equipment, if more than 4 hours elapse before their assembly into ignition elements, the units shall be dried at +100°F to +120°F for a minimum of 4 hours immediately prior to their being assembled into ignition elements.

3.6.2 Insulation resistance. Prior to assembly of the bridge-wire, the insulation resistance between the metal parts of the Inner Cup and Electrode Assembly, Drawing 1825936, shall be a minimum of 50 megohms when measured in accordance with 4.7.2 at a relative humidity of 80% or less.



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3.6.3 Bridgewire. The bridgewire shall be soldered as specified on Drawing 1825935.

3.6.3.1 Bridgewire assembly continuity. The bridgewire shall glow when subjected to the continuity test of 4.7.3.

3.6.4 Igniter assembly, loading. The consistency of the ignition charge and the loading procedure shall be regulated to secure an ignition charge which is free from internal voids while avoiding damage to the bridgewire. The ignition charge shall fill the cavity such that the charge is flush with the top of the cavity and the bridgewire is completely covered. All external surfaces of the Igniter Assembly, Drawing 1825934, shall be free from any of the ignition charge. Where necessary, external surfaces may be cleaned of ignition charge by means of a cloth wet with butyl acetate. The charged igniter assembly shall be dried at a temperature of +100°F to +120°F for a minimum of 12 hours. After removal of the units from the drying equipment, if more than 4 hours elapse before their assembly into ignition elements, the units shall be dried at +100°F to +120°F for a minimum of 4 hours immediately prior to their being assembled into ignition elements.

3.6.4.1 Ignition charge. The ignition charge shall be compounded by a method which maintains the weight composition of nonvolatile constituents prescribed on Drawing 1558439. The finished ignition charge when ready for buttering into the igniter assembly shall be a uniformly mixed slurry in which all of the particles are wetted. The consistency shall be such that the liquid will not separate from the mixture when the dish is tilted. Adjustments to the consistency may be obtained by the addition of pure butyl acetate or by evaporation of the solvent. (An advisory procedure for preparing and loading the ignition charge is given in 6.5.)

3.7 Ignition element inspection.

3.7.1 Visual inspection. Ignition elements shall be free of the following visible defects: dents, cracks, burrs, chips, improper crimp, split or cracked lip in the crimped area, perforated closure disc, incomplete or improper sealant application, or any other imperfections which could adversely affect performance or assembly into cartridges. See 4.7.4.

3.7.2 X-ray. Ignition elements shall be free of imperfections in assembly when examined by X-ray in accordance with 4.7.5.

3.8 Loaded spacer and propellant charges.



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3.8.1 Quantity. All loaded spacer and propellant charges shall be as specified on Drawings 2838286 and 2838297, respectively. Charges shall be determined by weight rather than by volume measurements.

3.8.2 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.8.2.1 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.6.

3.9 Cartridge inspection.

3.9.1 Visual inspection. Cartridges shall be free of the following visible defects: dents, deep scratches, sharp edges, cracks, burrs, split or cracked lip in the crimped area, damaged closure disc, incomplete or improper sealant application, or any other imperfections which could prevent entry into the firing chamber or adversely affect performance or resistance to moisture. See 4.7.7.

3.9.2 Bridge circuit resistance. The resistance of the bridge circuit in completely assembled cartridges, when measured electrically as specified in 4.7.8, shall be not less than 0.08 ohm nor more than 0.15 ohm.

3.9.3 Leakage. Cartridge leakage rate shall not exceed $10^{-5}$ cc of air or gas per second when tested in accordance with 4.7.9.

3.10 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.10.1, shall meet the requirements listed below when subjected to the test prescribed in 4.7.10.1.

3.10.1 Chamber pressure. The lower (L) limit of the maximum chamber pressure shall be 1200 psi when fired in the ejector rack of 4.7.10.1.

3.10.2 Velocity. The upper (U) limit for velocity of the store shall be 12.5 ft/sec. Velocity shall be measured over the interval from 6.4 to 12.4 inches of travel when fired in the ejector rack of 4.7.10.1.

3.10.3 Misfire. There shall be no misfires. See 4.7.10.2.3.

3.10.4 Hangfire. There shall be no hangfires. See 4.7.10.2.4.



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3.11 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414 or MIL-STD-105, whichever applies.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-414 or MIL-STD-105, whichever applies.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 76 MARK 51 MOD 1 impulse cartridges shall be prepared. Of these cartridges, 66 shall be expended in the tests listed below and 10 shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.6 through 3.9.3 of this specification. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: 15 "g" shock (6), vibration (6), 40-foot drop (6), and temperature and humidity cycling (12) except that 14 days shall be substituted for 28 days. The number in parentheses denotes the number of cartridges allocated for test. These cartridges (with exception of the 40-foot drop cartridges), after exposure to environmental treatments, together with 12 untreated



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cartridges are to be conditioned at +70° ± 5°F and fired as provided in 4.7.10.1. In addition, 24 cartridges shall be temperature conditioned for a minimum of 12 hours (12 at -65° ± 5°F and 12 at +160° ± 5°F) and fired as provided in 4.7.10.1. If any cartridge produces results outside the limits set forth in 3.10.1 through 3.10.4 or if any cartridge fails to fire, the preproduction sample shall be rejected. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 801 and 3200 cartridges, respectively. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 2838297, and verification of the following requirements and tests:



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| Item | Nature of Requirement Verification | Applicable Paragraphs | Classification |
|---|---|---|---|
| Primary Component | Single lot | 3.4, 4.3.3 | (M101) |
| Ignition Element | Age and acceptance | 3.6 | (C1) |
| Primer Composition | Moisture content | 3.6.1, 4.7.1 | (C2) |
| Loaded Spacer | Drying | 3.6.1 | (M102) |
| Inner Cup and Electrode Assembly | Insulation resistance | 3.6.2, 4.7.2 | (M103) |
| Bridgewire | Soldering | 3.6.3 | (C3) |
| Bridgewire Assembly | Continuity | 3.6.3.1, 4.7.3 | (C4) |
| Igniter Assembly | Loading | 3.6.4 | (C5) |
| Igniter Assembly | Drying | 3.6.4 | (M104) |
| Ignition Element | Visual inspection | 3.7.1, 4.7.4 | (M105) |
| Ignition Element | X-ray | 3.7.2, 4.7.5 | (C6) |
| Loaded Spacer and Propellant Charges | Type and weight | 3.8.1 | (C7) |
| Smokeless Powder | Condition | 3.8.2.1, 4.7.6 | (C8) |
| Cartridge | Visual inspection | 3.9.1, 4.7.7 | (M106) |
| Cartridge | Bridge circuit resistance | 3.9.2, 4.7.8 | (C9) |
| Cartridge | Leakage | 3.9.3, 4.7.9 | (C10) |
| Cartridge | Ballistic requirements | 3.10, 4.7.10 | (C11) |
| Cartridge | Packaging, packing and container markings | 4.6.1 | (M107) |

4.6.1 Packaging, packing and container marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Determining moisture content. Transfer approximately 2.0 grams of the primer composition to be used to a tared weighing dish, weigh to the nearest 0.1 milligram, calculate the weight of the sample. Transfer weighing dish and sample to a vacuum drying oven, dry at +71° ± 2°C and 2.0 ± 0.5 inches of mercury below atmospheric pressure until constant weight is obtained. The loss of weight is calculated as the percentage of moisture in the composition to be used.



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4.7.2 Insulation resistance. Prior to assembly of the bridgewires, the insulation resistance shall be checked with an applied voltage of 500 ± 25 volts DC. This measurement shall be made between the electrode and the inner cup. Units which fail to meet the requirements of 3.6.2 shall be rejected.

4.7.3 Bridgewire assembly, continuity. The assembly shall be thoroughly cleaned to remove flux and contamination. Subsequent to cleaning, the bridgewires shall be flashed by application of an electric current to assure continuity of both bridges. The current shall be regulated to produce a visible red glow of the bridges. An electromotive force of approximately 1.5 volts AC or DC may be used with a suitable resistor to adapt the source characteristics to the bridge resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both bridges shall be rejected. The faulty assemblies may be reworked by replacement of the bridges.

4.7.4 Ignition element visual inspection. Examine all ignition elements and reject those having any of the defects of 3.7.1.

4.7.5 Ignition element X-ray. Ignition elements shall be positioned on their side for the most revealing exposure. Examine all ignition elements by X-ray and reject those having observable imperfections in assembly such as: missing, defective or improperly assembled components; charges missing or abnormally small; or any other imperfections which could adversely affect performance.

4.7.6 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable portions of paragraphs 1308-1318 and 1322 of OP 5. Evidence of decomposition shall be cause for rejection.

4.7.7 Cartridge visual inspection. Examine all completely assembled cartridges and reject those having any of the defects of 3.9.1.

4.7.8 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.9.2 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.9.2 shall be rejected. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.



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4.7.9 Leakage. Each cartridge in the preproduction sample or ballistic sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of $10^{-5}$ cc/sec gas or air at a pressure differential of one atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the ballistic sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.10 Ballistic tests.

4.7.10.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Tables A-2 and B-1 of MIL-STD-414, Inspection Level IV and an AQL of 0.10. The cartridges shall be conditioned for a minimum of 12 hours at +70° ± 5°F and fired in a horizontally mounted Aero 7A ejector rack. A cylindrically shaped store with the following characteristics shall be suspended on the ejector rack:

a. Weight - 500 pounds

b. Length - 10 feet

c. External diameter - 10.75 inches

d. Distance between suspension lugs - 14 inches

e. Location of center of gravity - 3.5 inches forward of after suspension lug

A Filler Plug, Drawing 2171790, shall be installed in the longer chamber of the Aero 7A rack. Suitable instrumentation for measuring chamber pressure, velocity and determining the time from the application of firing current to hook opening shall be used to determine if the acceptance criteria are met. The firing current shall be 10 ± 0.5 amperes DC. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from the production lot for investigative purposes.

4.7.10.2 Criteria of acceptability.

4.7.10.2.1 Chamber pressure. The lower (L) limit of maximum pressure shall be as specified in 3.10.1. With respect to chamber pressure, the lot shall be acceptable if the quantity $(\bar{X}-L)/s$ is equal to or greater than the appropriate value of k, at an AQL of 0.10, when computed in accordance with MIL-STD-414, Part I, Table B-1.



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4.7.10.2.2 Velocity of the store. The upper (U) limit of store velocity shall be as specified in 3.10.2. With respect to store velocity, the lot shall be acceptable if the quantity $(U-\bar{X})/s$ is equal to or greater than the appropriate value of k, at an AQL of 0.10, when computed in accordance with MIL-STD-414, Part I, Table B-1.

4.7.10.2.3 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.7.10.2.4 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between application of current and hook opening time exceeds 0.030 second.

4.7.11 Retest. There shall be no retests.

4.7.12 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and the complete test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Seven cartridges shall be packaged in an inner container in accordance with LD 534542.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116, and with the Code of Federal Regulations 49 CFR 71-78.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Eight inner containers as described in 5.1.1 shall be packed in accordance with LD 537905.

5.2.2 Level B. Not applicable.



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5.2.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests, and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on the drawings of LD 534542 and LD 537905, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 51 MOD 1 impulse cartridge is intended for use in Navy bomb ejector racks.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required, and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.



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6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Levels A and C packaging and packing given below were extracted from BUSANDA Instruction 4030.17.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations, and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in DOD 4145.26M, "DOD Contractors' Safety Manual for Ammunition, Explosives and Related Danagerous Material".

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